

Q.1.

1. What is algorithm?



An algorithm is a set of well-defined instructions or a step-by-step procedure designed to perform a specific task or solve a particular problem. In computing, algorithms are fundamental to programming and data processing, as they provide a structured approach to executing tasks efficiently.

Here are some key aspects of algorithms:

1. **Defined Steps:** An algorithm consists of a clear, finite sequence of steps that must be followed in order to achieve the desired outcome.
2. **Input and Output:** An algorithm typically takes input (data or parameters) and produces output (results or solutions).

3. **Unambiguity:** Each step in an algorithm must be precisely defined so that there is no ambiguity in how the steps are to be executed.
4. **Termination:** An algorithm must eventually terminate after a finite number of steps, producing a result or completing the task.
5. **Efficiency:** While not always required, an efficient algorithm performs its task with the least amount of resources (time, space, etc.) necessary.

For example, a simple algorithm for adding two numbers might be:

1. Start
2. Take two numbers as input (let's call them A and B)
3. Add A and B
4. Output the result
5. End

2. What is flowchart?



A flowchart is a visual representation of a process or algorithm. It uses symbols and arrows to illustrate the sequence of steps involved in completing a task or solving a problem. Flowcharts are valuable tools for designing, analyzing, and communicating processes because they provide a clear, graphical way to represent complex procedures.

Here are some common elements of flowcharts:

1. **Start/End Symbols:** These are usually represented by ovals or rounded rectangles and indicate the beginning and end of a process.
2. **Process Steps:** Represented by rectangles, these symbols denote specific actions or operations that need to be performed.
3. **Decision Points:** Shaped like diamonds, these symbols represent points in the process where a decision must be made, leading to different paths based on the outcome (e.g., "Yes" or "No").
4. **Arrows:** These show the direction of flow and connect different steps, illustrating the sequence in which actions are performed.

5. Input/Output Symbols: Parallelograms are used to represent data inputs and outputs.

6. Sub-processes: Represented by rectangles with double-struck vertical edges, these symbols indicate complex steps that are defined in separate flowcharts.

Here's a simple example of a flowchart to illustrate how to make a cup of tea:

1. Start

2. Boil Water (Process Step)

3. Is Water Boiled? (Decision Point)

- **Yes:** Move to the next step
- **No:** Return to the "Boil Water" step

4. Add Tea Bag to Cup (Process Step)

5. Pour Water into Cup (Process Step)

6. Steep Tea (Process Step)

7. Remove Tea Bag (Process Step)

8. Add Sugar/Milk? (Decision Point)

- **Yes:** Add sugar/milk (Process Step)
- **No:** Skip to the next step

9. **Stir Tea** (Process Step)

10. **End**

Flowcharts are useful in various contexts, including software development, business process management, and system design, as they help clarify the steps involved and identify potential improvements.